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Executive Summary

Deep-sea gold (Au) and silver (Ag) are mainly hosted in **seafloor massive sulfide (SMS) deposits** and, to a lesser extent, in **ferromanganese crusts** and **placer-like sediments**. Hydrothermal vents (mid-ocean ridges, back-arc basins, volcanic arcs) spawn sulfide mounds that can carry several grams of Au and kilograms of Ag per tonne【35†L226-L233】【85†L1-L4】. Cobalt-rich crusts on seamounts concentrate base metals and occasionally Ag–Au (e.g. SW Pacific crusts with "unusual" Ag–Au enrichment【40†L61-L65】). Polymetallic nodules (Ni–Cu–Co–Mn) have very low Au/Ag, so are minor targets. We identify ~10 **high-probability regions**, e.g. Western Pacific back-arcs (recent Chinese find), Red Sea rift, Tonga-Kermadec-Lau arcs, East Pacific Rise, Mid-Atlantic Ridge vents, large seamount fields (Pacific cobalt crusts), major abyssal nodule plains (Clarion-Clipperton, Central Indian), and sedimentary fans (e.g. Bengal). Table 1 ranks these by geology, evidence, depth and access (see below). Recommended exploration uses multibeam/magnetic surveys, plume geochemistry, and targeted ROV/drill sampling. Environmental and legal considerations (deep-ecosystem sensitivity; UNCLOS/ISA regimes) are critical. Major data gaps include sedimentary Au concentrations and high-res mapping of continental-margin fans.

【15†embed_image】 *Figure: Global distribution of known hydrothermal vent fields (red = confirmed, yellow = inferred)【35†L214-L220】. Many are on mid-ocean ridges and back-arc systems – prime Au/Ag targets.*

Seafloor Deposit Types and Settings

- **Polymetallic Sulfides (Hydrothermal Vents):** SMS form where hot, mineral-rich fluids exit the seafloor. These are overwhelmingly the richest Au/Ag sources. Vent chimneys and mounds precipitate chalcopyrite, sphalerite, pyrite and other sulfides. In early high-T stages (350–400 °C), anhydrite and chalcopyrite dominate; later cooler stages (200–300 °C) precipitate Zn-sulfides and Au/Ag minerals. Gold occurs as native Au and electrum, often with tellurides (e.g. hessite Ag_2Te, altaite PbTe)【85†L1-L4】. Silver is typically ubiquitous as a trace in Cu–Zn ores (e.g. marmatite-(Ag) or argentite in gossans). Deposits are found on **mid-ocean ridges**, **back-arc basins**, and **island arcs**【35†L214-L220】. Known high-Ag/Au fields include the Red Sea rift and Eastern Pacific Rise【40†L61-L65】. For example, Chinese surveys recently reported a Western Pacific back-arc vent field (within China's EEZ) with ore grading ~15 g/t Au and >1 kg/t Ag【35†L226-L233】.

- **Cobalt-rich Ferromanganese Crusts:** These Fe–Mn oxide crusts encrust seamount flanks (600–2500 m depths), grown slowly from seawater【52†L323-L332】. They are rich in Co, Ni and REE, with minor Cu/Pt/Pd. Precious metals are usually low, but isolated *Au/Ag anomalies* exist. An early report found "unusual" Au and Ag enrichments in SW Pacific back-arc crusts【40†L61-L65】. More commonly, trace Au in crusts is associated with mixed hydrogenetic/hydrothermal inputs. Thick crusts are mostly on Pacific seamount chains within Pacific-island EEZs【52†L331-L339】; Atlantic and Indian occurrences are sparse.

- **Polymetallic Nodules:** Ni–Cu–Co–Mn nodules blanket abyssal plains (e.g. Clarion-Clipperton, Peru Basin, Central Indian Ocean) at ~3500–6000 m depth【25†L198-L205】. They mainly host nickel, copper, cobalt and manganese. Gold and silver are extremely minor (enrichment factors $\lesssim$1000× crustal for Au in nodules have been reported【40†L52-L60】, but absolute grades are

very low). Thus nodules are not prime Au/Ag targets, though they mark regions of deep, sediment-starved seabed – a baseline for other searches.

- **Sediment-hosted "Placer" Deposits:** Analogous to onshore heavy-mineral placers, these would occur where rivers have delivered Au/Ag (often bound to heavy minerals) into canyons, fans or shelf deposits. Examples might include large submarine fans (Bengal, Indus, Congo) or submerged paleo-placer sands. Data are very scarce, but coarse sulfidic sands have been dredged in places (e.g. off Namibia) with base metals and minor Au. We note these as *exploratory possibilities*, emphasizing continental slopes and fans near known mineral-bearing hinterlands.

Geochemical and Mineralogical Indicators

Geochemically, Au and Ag in seafloor deposits often correlate with **chalcophile and tellurium** elements. For sulfides, elevated As, Sb, Bi, Te can signal precious-metal halos. Hydrothermal fluids carrying Au are "fisher-troated" by sulfur; Au often precipitates late when sulfide saturation is reached. High Cu/S ratio fluids may also favor Au deposition. In SMS, native Au and Ag are found with chalcopyrite cores or in quartz/opal infill; electrum often cements chalcopyrite and sphalerite fractures 【85†L1-L4】. Tellurides (e.g. hessite, sylvanite $AgAuTe_4$) are common Au/Ag minerals in mature chimneys. In crusts, Au often occurs as tiny grains or alloys within Mn oxides or accessory sulfides. **Paragenetic sequence (in SMS)** typically is: anhydrite → chalcopyrite + pyrite → sphalerite + galena → barite/Fe-oxyhydroxides, with Au/Ag intergrown during sulfide growth or late silica cementation 【85†L1-L4】. Thus, coincident Cu–Zn–Pb–(Au+Ag) and Te–Bi anomalies are strong Au/Ag indicators in geochemical surveys.

Known Occurrences and High-Potential Regions

Many vent fields worldwide contain trace Au/Ag【35†L226-L233】. Notable examples:

- **East Pacific Rise (Galápagos Rift)** – Known for high-grade vents (e.g. Alarcon Rise) with Au (up to 10s of g/t in some mounds).
- **Mid-Atlantic Ridge (TAG, Rainbow)** – Slow-spreading ridge vents with some Au.
- **Back-arc Basins** – Tonga–Kermadec–Lau (South Pacific) and Mariana (North Pacific) arcs have several gold-bearing fields.
- **Red Sea Rift** – Hydrothermal brines deposit barite–sulfides with reported Au/Ag; e.g. Shaban brine pools show Au in solution【40†L61-L65】.
- **Western Pacific Arcs** – Recent Chinese surveys report a rich gold-silver sulfide field in a back-arc basin (within China's EEZ)【35†L226-L233】.
- **Cratonic Continental Margins** – (Caspian Sea muds, Gulf of California exhalites not covered here) but shallow.
- **Cobalt Crust Seamounts** – Pacific seamounts (Emperor-NW Pacific, Micronesian plateaus) have occasional Au-bearing crust pieces.

We compiled candidate regions by overlaying these geological provinces. The WHOI vent map (Fig.1) shows vent distributions; we highlight vent- and seamount-rich zones (SW Pacific, NW Pacific, MAR, EPR, Indian ridges) as priority areas. Continental-slope fans (e.g. Bengal Fan off Bangladesh/India; Indus Fan; Zambezi Fan) are also flagged as unknown but intriguing.

Top 10 Target Areas (Ranked)

Rank	Target Area (Lat, Lon ±)	Deposit Type	Key Points (depth, geology, evidence, legal)
1	**Western Pacific back-arc** (≈20°N, 130°E ±2°)	SMS (hydrothermal)	Recent Chinese expedition discovered Au–Ag-rich sulfide mounds (up to ~15 g/t Au, >1 kg/t Ag)【35†L226-L233】. Likely a back-arc basin (>700 m depth), 350–400 °C vents. High-grade Au/Ag. China's EEZ (clear national jurisdiction) ensures permitting; depth ~1–1.5 km (reachable by ROV). Geological setting: active arc/back-arc (extensional) – prolific for Au【85†L1-L4】.
2	**Red Sea rift (Axial trough)** (≈22°N, 38°E ±0.5°)	SMS (hydrothermal)	Shallow (~200–2300 m) hydrothermal field on the Gulf of Suez rift. Known for Zn–Pb–Ag–Au in brines【40†L61-L65】. Reports of significant Au/Ag (e.g. 8–12 g/t Au) in Red Sea ores. Located in Egyptian/ Saudi EEZ; easier logistic (ship-based submersibles). Actively spreading ridge in anoxic environment (high Au/Ag potential【40†L61-L65】).
3	**Mid-Atlantic Ridge (Rainbow/ TAG)** (≈36°N, 33°W; ≈26°N, 44°W)	SMS (ridge)	Deep (~2300–3600 m) black-smoker fields. Rainbow and TAG complexes have known Au (ppm–ppb) and high Ag (e.g. supergene Ag in gossans). International waters (ISA "Area") with high legal uncertainty (no exploitation code yet). Low-temperature base-metal-rich vent setting; moderate Au, but global significance; accessibility by deep-ROV.
4	**Kermadec Arc (e.g. Brothers Island)** (≈30°S, 179°W ±2°)	SMS (arc/back-arc)	Active vents at ~1600 m (Brothers, Macauley, etc.). Known Au (~10 g/t) and Ag-rich chimneys. Located in New Zealand EEZ (clear regulations). Complex magmatic arc – high Au geochemistry (similar to onshore Au fields). Depth ~1–2 km.
5	**Lau Basin / Tongan arc** (≈20°S, 175°W ±1°)	SMS (back-arc)	Abundant hydrothermal fields (Tui Malila, Niua South, etc.) at ~1000–1500 m. VMS with reported Au up to ~5–10 g/t, high Ag. Located in Tongan/ Niuean/EU (EUR-OPA) waters; complex jurisdiction (some ROP-EU cooperation). Young, hot back-arc setting (very high heat flow). Depth ~1–2 km.
6	**East Pacific Rise (Alarcon/ Galápagos Rift)** (≈2°S, 91°W ±0.5°)	SMS (ridge)	Fast-spreading ridge vents at ~2500–3000 m. Known for gold-rich sulfides (e.g. 5–20 g/t Au in some chimneys). Within international (but Ecuadorian) EEZ. High magma supply – rich metals. Challenging depth ~3 km but many expeditions conducted.

Rank	Target Area (Lat, Lon ±)	Deposit Type	Key Points (depth, geology, evidence, legal)
7	**Clarion-Clipperton Abyssal Plain** (≈12°N, 119°W ±2°)	Nodules (abyssal)	Vast nodule field at ~4000–5000 m depth (international waters, ISA Area). Nodules contain Ni/Cu/Co, trace Au/Ag (enriched 50–100× crust for Ag, 100–1000× for Au in nodules【40†L52-L60】, but absolute grades ~ppb). Accessibility: very deep (4–5 km), high-pressure; tech possible (ROV/AUV mining trials exist). Only ISA regulation zone (exploration contracts active).
8	**Central Indian Basin (India EEZ)** (≈5°S, 90°E ±2°)	Nodules (abyssal)	Known nodule area (1996 India contract). 4500–5500 m deep. Similar comment as CCZ: Ni–Cu–Co resource, only trace Au/Ag. Indian EEZ offers sovereignty (India marine policy), but extreme depth. High sedimentation (May affect nodule grade).
9	**NW Pacific Seamounts (Emperor chain)** (≈40°N, 172°W ±5°)	Crusts (seamount)	Old Pacific volcanic seamounts in Japanese/US EEZ. Crusts 800–2500 m deep, thickest on older seamounts. Contains Co, Ni and trace Au (visible Au grains reported in similar Pacific crusts【40†L61-L65】). Accessibility: 1–3 km depth (ROV feasible), Japan/Russia-US EEZ (clear jurisdiction, but often remote). Data: known crust projects by JAMSTEC, Japan Oil Minerals.
10	**Bengal Fan (Bay of Bengal)** (≈15°N, 90°E ±2°)	Placer-like (fan)	Deep-marine fan off Bangladesh/India (depth ~4000–6000 m). Potential to host dense heavy minerals (Au, Ti, Zr) from Himalayan rivers. Very little exploration data exist. Bangladesh/India EEZ (Jurisdiction clear). Survey depth ~4–6 km (challenging). Sediment geochemistry and sampling recommended.

Table 1. Top 10 high-Au/Ag deep-sea target areas by geology, evidence, depth and legal status.

Each target warrants tailored surveys: e.g. bathymetric swath mapping, magnetics/gravity (to delineate structures), and water-column plume sampling (to detect vent effluents). Confirmatory sampling (dredge, corer, ROV) is next. See Section **Survey Strategy** for details.

Exploration Methods

A multi-stage workflow is standard:

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flowchart LR
    A[Pre-survey planning: Compile bathymetry & existing data] -->
B[Geophysical surveys: Multibeam bathymetry, sidescan, magnetics, EM]
    B --> C[Water-column sampling: Plume-tracking (CHEMINI/AUV) and
geochemical analysis]
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C --> D[Focused mapping: High-res bathymetry & sub-bottom profiler]
D --> E[Ground truth sampling: ROV/AUV video, rock coring, dredging]
E --> F[Laboratory analysis: Mineral assays (Au/Ag), SEM/EM of Au phases]
F --> G[Resource evaluation & modeling]
G --> H[Environmental baseline: Bio-survey, geotech sampling]
H --> I[Permitting & impact assessment]
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- **Remote sensing and Geophysics:** Initial reconnaissance uses satellite altimetry (for gravity anomalies), public bathymetry grids, and commercial data to identify ridges, seamounts, or fans. Shipborne multibeam sonar maps seafloor morphology; sidescan can highlight unusual textures (e.g. vents). Magnetics/EM detects conductive ore bodies (massive sulfides are magnetic and highly conductive).
- **Geochemical surveys:** Deep-towed or AUV-mounted sensors (e.g. AND/ORP, turbidity) locate plumes of vent particles. Water sampling (e.g. CTD-mounted chemosamplers) can identify metal anomalies (Fe, Mn, Ba) indicative of vents. These "hotspot" markers focus search areas.
- **Sampling:** ROVs or HOVs collect chimney fragments, crust cobbles or sediment cores. Box cores and grabs sample nodules or heavy minerals. Cores through vents or fans help delineate extent. Subsurface drilling (e.g. ROV drills) can access ore at depth.

Advances in AUVs and deep-sea robots now allow systematic high-resolution surveys of large areas. For example, AUVs with bathy/laser systems can map km² patches of seamounts. Deep-towed magnetometer systems can "map" sulfides under sediment.

Figure 2: *Example workflow for mineral exploration (see mermaid flowchart above).*

Environmental & Legal Considerations

Deep-sea mining raises major environmental issues. Hydrothermal vents and seamounts host unique ecosystems; disturbance can cause long-lasting impacts (loss of microbial colonies, benthos). Nodules and sediment plumes can smother seafloor fauna. Many target areas lack baseline biological data. Environmental Impact Assessments (EIAs) and the ISA's Regional Environmental Management Plans (REMPs) will govern any activity【35†L214-L220】.

Legally, seabed beyond national jurisdictions ("the Area") falls under the **ISA**. Currently only exploration (not mining) is permitted under UNCLOS. Notable points: Clarion-Clipperton and CCZ are in the Area (multiple exploration contracts exist【76†L207-L215】), so a future moratorium or strict regulations could apply. Within EEZs, national laws (e.g. Papua New Guinea, Cook Islands, India) prevail; some nations (e.g. Tonga) have already begun drafting mining policies. The Western Pacific Chinese find lies within China's EEZ, avoiding ISA ambiguity. Red Sea is national waters. Thus legal/accessibility varies: **Priority sites in EEZs** (China, NZ, Tonga, India) are easier to permit; high-seas sites (CCZ, MAR) require complex agreements.

Environmental safeguards (no-go zones, avoidance of biologically sensitive areas, monitoring of sediment plumes) must be integral to any survey. Technologies should minimize seabed contact (e.g. suction mining vs. bucket grabs). Preliminary animal and geochemical baselines are urgent data gaps.

Survey & Sampling Strategy per Target

- **Hydrothermal Fields:** Use high-resolution bathymetry + magnetics to map vent topography. AUV plume surveys to find active vents. ROV dives to collect chimney and wall rock for Au/Ag

assays (SEM/TEM to identify Au phases). Core through sulfide mounds to measure grade distribution.
- **Cobalt Crusts:** Map seamount chains using multibeam; identify flat flanks suitable for crust growth (800–2000 m). Drilled cores or ROV grabs of crust to assay Au and Co. Radiometric dating of crust layers to estimate growth rates.
- **Nodule Fields:** Wide-area AUV/ROV nodules counts with camera, combined with box cores for grade. Bulk mining trials (as in Clarion-Clipperton demos) to test retrieval. Minimal need for Au assay (due to low grades).
- **Sediment Fans:** Ship-mounted echosounder to delineate channel/fan architecture. Grab sampling or gravity cores along supposed heavy-mineral channels. Geochemical screening (DGT probes, magnetic separation) for heavy minerals. This is highly exploratory—likely starting with the Bengal Fan as largest known submarine fan.

Each survey should integrate environmental observation (video of fauna, sediment composition). We suggest at least two-phase survey: remote sensing first, then focused in-situ measurement with ROVs/AUVs. All samples need careful provenance and chain-of-custody logging.

Data Gaps & Research Priorities

- **Sedimentary Gold:** Virtually no data exist on gold in major deep-sea fans. Priority: trace-element analyses of sediments from Bengal, Indus, Amazon fans.
- **Mineral Paragenesis:** Better petrological models of Au/Ag deposition at vents (e.g. fluid-inclusion studies) to refine exploration vectors.
- **Crust Geochemistry:** More reconnaissance of cobalt crusts for Au/Ag; characterize trace Au in crust growth zones.
- **High-Resolution Mapping:** Many continental margins and seamount regions lack modern bathymetry. Uncharted geomorphology hinders target identification.
- **Environmental Baselines:** Essential: document vent/seamount ecosystems before any mining.

Addressing these gaps will improve targeting and environmental management. Collaboration across countries (open data) will be key.

Sources: See references cited above (peer-reviewed and official reports) for data on deposit grades, global occurrences and resource estimates【25†L198-L205】【35†L214-L220】【40†L61-L65】【52†L331-L339】【85†L1-L4】.
